



Web Fundamentals

WEF1VO

Teil 12: HTML JavaScript APIs

Medientechnik und -design | WS 2025/26

HTML JavaScript **APIs**

Canvas, Video, Audio, Web
Storage, Geolocation, Drag & Drop
uvm.

HTML-Technologien mit Zugriff
über JavaScript.

Auch einfach **Web APIs** genannt.

HTML





Browser Kompatibilität

Can I Use...

<https://caniuse.com/>

Unterstützungsmatrix von Features für alle
Browser

Feature Detection

```
if ("geolocation" in navigator) {  
    // Geolocation ist verfügbar  
} else {  
    // Fallback oder Fehlermeldung  
}
```

Canvas

Eine auflösungsabhängige Bitmap-Zeichenfläche.



The canvas element provides scripts with a resolution-dependent bitmap canvas, which can be used for rendering graphs, game graphics, art, or other visual images on the fly.

Canvas anlegen



HTML-Datei:

```
<canvas width="640" height="480" id="draw">
```

```
  Ihr Browser unterstützt Canvas leider nicht!
```

```
</canvas>
```

JS-Datei:

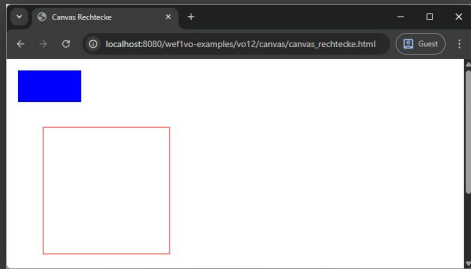
```
const canvas = document.getElementById("draw");
```

```
const ctx = canvas.getContext("2d");
```

Rechtecke zeichnen



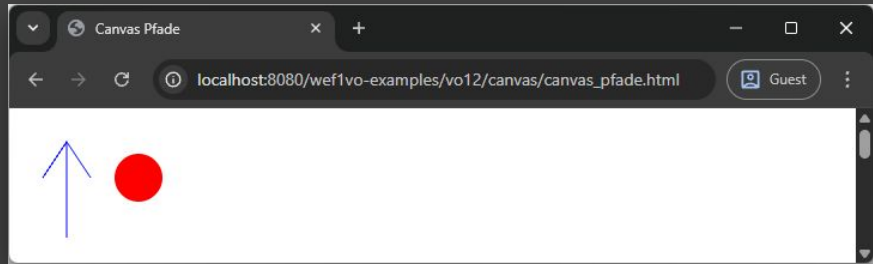
```
ctx.fillStyle = "blue";  
ctx.strokeStyle = "#FF0000";  
ctx.fillRect(10, 10, 100, 50);  
ctx.strokeRect(50, 100, 200, 200);
```



Pfade zeichnen



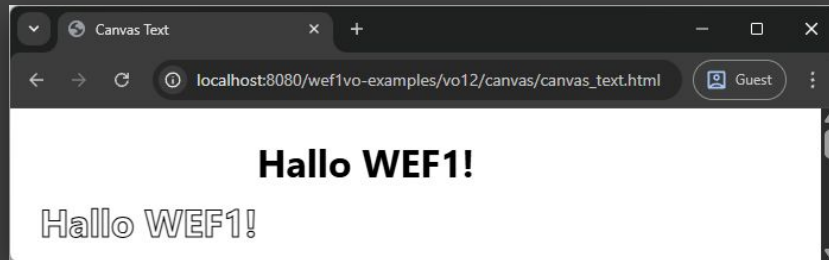
```
ctx.strokeStyle = "blue";
ctx.moveTo(20, 50);
ctx.lineTo(40, 20);
ctx.lineTo(60, 50);
ctx.moveTo(40, 20);
ctx.lineTo(40, 100);
ctx.stroke();
ctx.beginPath(); // Beginne neuen unabhängigen Pfad
ctx.fillStyle = "red";
ctx.arc(100, 50, 20, 0, 2 * Math.PI)
ctx.fill();
```



Text darstellen



```
ctx.font = "bold 24pt system-ui, sans-serif";  
ctx.fillText("Hallo WEF1!", 200, 50);  
ctx.textAlign = "right";  
ctx.strokeText("Hallo WEF1!", 200, 100);
```



Bilder darstellen



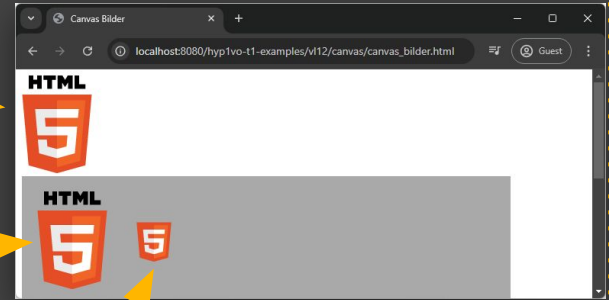
```

```

```
const img = document.getElementById("image");
ctx.drawImage(img, 20, 20);
```

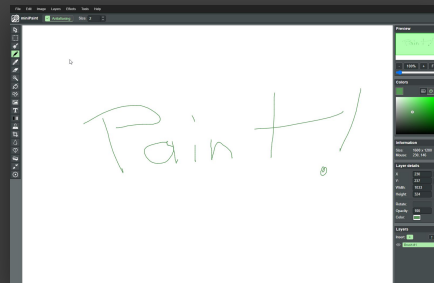
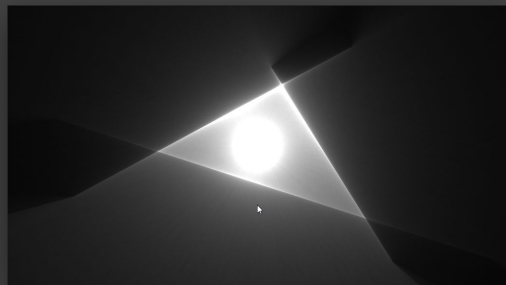
```
const jsImg = new Image();
jsImg.src = "html5logo.png";
jsImg.addEventListener("load", function () {
  ctx.drawImage(jsImg, 0, 24, 92, 104, 150, 60, 46, 52);
});
```

Diagram: Three yellow arrows point from the code to the browser. The first arrow points from the `id="image"` attribute to the first logo. The second arrow points from the `drawImage(img, 20, 20)` call to the first logo. The third arrow points from the `drawImage(jsImg, 0, 24, 92, 104, 150, 60, 46, 52)` call to the second logo.





Canvas Beispiele



Video

Videos ohne Plugin im Browser abspielen



Container & Codecs

Format	Chrome 144	Firefox 147	Edge 144	Opera 125	Safari 26.2
MP4 (.mp4): H.264 + AAC/MP3	•	○/•	○	•	•
MP4 (.mp4): H.265 + AAC	○	○	○	○	•
MP4 (.mp4): AV1 + AAC/Opus	•	○/•	•	•	•
WebM (.webm): VP8 + Vorbis	•	•	•	•	•
WebM (.webm): VP9 + Opus	•	•	•	•	•
WebM (.webm): AV1 + Opus	•	•	•	•	•

Video im Browser



Einzel:

```
<video src="../../assets/oceans-clip_vp9_opus.webm"
       width="640" height="264" controls>Fallback-Text</video>
```

Mehrere:

```
<video width="640" height="264" controls>
  <source src="../../assets/oceans-clip_av1_opus.webm" type="video/webm">
  <source src="../../assets/oceans-clip_av1_opus.mp4" type="video/mp4">
  <source src="../../assets/oceans-clip_vp9_opus.webm" type="video/webm">
  <source src="../../assets/oceans-clip_h265_aac.mp4" type="video/mp4">
  <source src="../../assets/oceans-clip_h264_mp3.mp4" type="video/mp4">
  <source src="../../assets/oceans-clip_vp8_vorbis.webm" type="video/webm">
</video>
```

Eigene Kontrollelemente

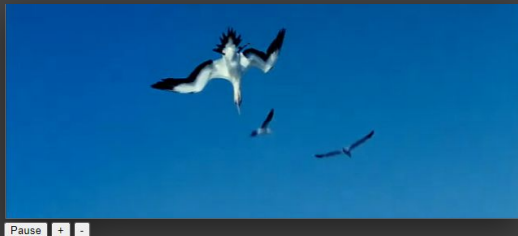


HTML-Datei:

```
<video width="640" height="264" id="vid">...</video>  
<button type="button" id="playpause">Play</button>
```

JS-Datei:

```
const vid = document.getElementById("vid");  
const playPause = document.getElementById("playpause");  
  
playPause.addEventListener("click", function () {  
    vid.paused ? vid.play() : vid.pause();  
    playPause.textContent = vid.paused ? "Play" : "Pause";  
});
```



Audio

Audio ohne Plugin im Browser abspielen



Container & Codecs

Format	Chrome 144	Firefox 147	Edge 144	Opera 125	Safari 26.2
WAVE (.wav): PCM	•	•	•	•	•
WebM (.webM): Vorbis	•	•	•	•	•
Ogg (.ogg): Vorbis	•	•	•	•	•
MP3 (.mp3): MP3	•	•	•	•	•
MP4 (.mp4): MP3	•	•	•	•	•
MP4 (.mp4): AAC	•	○	•	•	•
FLAC (.flac): FLAC	•	•	•	•	•
Ogg (.ogg): FLAC	•	•	•	•	•

Audio im Browser



Einzel:

```
<audio src="../../assets/miaow-bubble.mp3" controls>Fallback-Text</audio>
```

Mehrere:

```
<audio controls>
  <source src="../../assets/miaow-bubble.flac" type="audio/flac">
  <source src="../../assets/miaow-bubble.mp3" type="audio/mpeg">
  <source src="../../assets/miaow-bubble.ogg" type="audio/ogg">
</audio>
```

Geolocation

Die Position von Nutzer*innen abfragen

Position ermitteln



```
navigator.geolocation.getCurrentPosition(showMap);
```

```
function showMap(position) {  
  const lat = position.coords.latitude;  
  const long = position.coords.longitude;  
  // Zeige eine Karte  
}
```

Karte mit Leaflet



Library: Leaflet (<https://leafletjs.com/>)

HTML-Datei:

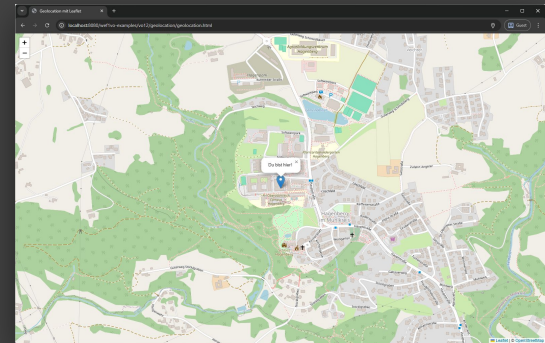
```
...  
<head>  
  ...  
  <link rel="stylesheet" href="https://unpkg.com/leaflet@1.9.4/dist/leaflet.css"  
    integrity="sha256-p4NxAoJBhIIN+hmNHzRCf9tD/miZyoHS5obTRR9BMV="   
    crossorigin="" />  
  <script src="https://unpkg.com/leaflet@1.9.4/dist/leaflet.js"  
    integrity="sha256-20nQCchB9co0qIjJZRGuk2/Z9VM+kNiyxNV1lvTLZBo="   
    crossorigin="" defer></script>  
  <script src="map.js" defer></script>  
</head>  
<body>  
  <div id="map"></div>  
</body>  
...
```

Karte mit Leaflet



JS-Datei (map.js):

```
function initialize() {  
    const map = L.map("map").setView([47.61, 13.783], 8);  
    L.tileLayer("https://tile.openstreetmap.org/{z}/{x}/{y}.png", {  
        maxZoom: 19,  
        attribution: '&copy; <a href="http://www.openstreetmap.org/copyright">OpenStreetMap</a>'  
    }).addTo(map);  
  
    if (navigator.geolocation) {  
        navigator.geolocation.getCurrentPosition(position => {  
            const lat = position.coords.latitude;  
            const lng = position.coords.longitude;  
            map.setView([lat, lng], 16);  
  
            L.marker([lat, lng]).addTo(map)  
                .bindPopup("Du bist hier!")  
                .openPopup();  
        }, () => { handleLocationError(true, map); });  
    } else {  
        handleLocationError(false, map);  
    }  
}
```



Local Storage

Daten lokal im Browser abspeichern

Post-It fürs Web



HTML-Datei:

```
<section id="sticky" contenteditable="true"></section>
```

In JavaScript: `localStorage`-Objekt:

- `setItem(key, value)`: Speichern
- `getItem(key)`: Auslesen
- `removeItem(key)`: Löschen

Post-It fürs Web

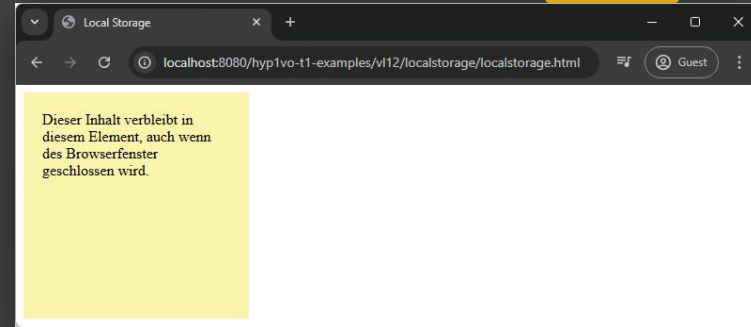


JS-Datei:

```
const defaultText = "Schreib auf mich!";
const note = document.getElementById("sticky");
const storedValue = localStorage.getItem("content");

if (storedValue) {
  note.innerHTML = storedValue;
} else {
  note.innerHTML = defaultText;
}

note.addEventListener("blur", (event) => {
  const plainText = event.target.textContent;
  if (plainText.trim() === "" || plainText === defaultText) {
    localStorage.removeItem("content");
    event.target.innerHTML = defaultText;
  } else {
    localStorage.setItem("content", event.target.innerHTML);
  }
});
```

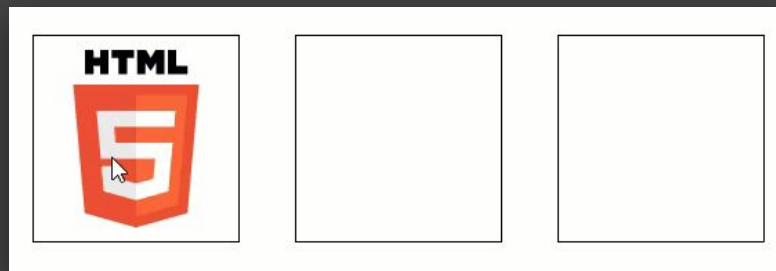


Drag & Drop

Elemente von A nach B ziehen



4 Schritte



1

Ziehbar machen

2

Mit dem Ziehen starten

3

Elemente zum
Fallenlassen
auswählen

4

Operationen beim
Fallenlassen ausführen

Ziehbar machen



```
<div class="dropzone">  
    
</div>
```

```
<div class="dropzone"></div>
```

```
<div class="dropzone"></div>
```

Restliche Schritte



```
const dragImage = document.getElementById("dragImage");  
const dropZones = document.querySelectorAll(".dropzone");
```

```
dragImage.addEventListener("dragstart", function (event) {  
  event.dataTransfer.setData("text", event.currentTarget.id);  
});
```

← Ziehen starten

```
dropZones.forEach(zone => {  
  zone.addEventListener("dragover", event => {  
    event.preventDefault();  
  });  
  zone.addEventListener("drop", event => {  
    event.preventDefault();  
    const data = event.dataTransfer.getData("text");  
    const draggedElement = document.getElementById(data);  
    event.currentTarget.appendChild(draggedElement);  
  });  
})
```

← Elemente zum
Fallenlassen auswählen

← Operationen beim
Fallenlassen durchführen

More APIs

Web Sockets, Web Workers, Service Workers,
Indexed Database uvm.





Klausur & Vorbereitung

Klausur-Hangout



Klausur

- e-Klausur in Moodle am Freitag, 06.02.2026.
 - 08:50–11:15 SUB & WHS.
- Stoff als PDF im eLearning-Kurs.
- Klausur-Hangout über MS Teams am Tag vorher?
z.B. Donnerstag, 05.02., um 17:00 Uhr?
 - Bitte intern abstimmen und bald Bescheid sagen!
 - Link zu Fragenportal in Kurs online.
 - Dort Fragen stellen, die beim Lernen auftauchen.
 - Werden dann im Livestream mit Beispielen besprochen.

Credits



- S. 2: W3C HTML5 Logo: <https://www.w3.org/html/logo/>